

Solution Requirements

Date	26 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	4Marks

Define Functional and Non-Functional Requirements:

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	 Authentication	Users (customers, loan officers, admin) shall be able to securely register, login, and verify their identity using email/mobile and OTP.	High
2	 Authorization (Access Control)	The system shall define roles (Customer, Loan Officer, Underwriter, Admin) and provide role-based access to features and data.	High
3	 Loan Application Management	Customers shall be able to submit loan applications, provide required details, upload documents, and save progress.	High
4	 Application Review & Approval	Loan officers shall review applications, validate documents, assess risk, and approve/decline loans with remarks.	High
5	 Credit Assessment & Scoring	The system shall integrate with credit bureaus, retrieve credit reports, calculate credit scores, and support decision making.	High
6	 Disbursement Management	Once approved, the system shall initiate loan disbursement via the payment gateway/bank and track status.	High
7	 Repayment Management	The system shall generate repayment schedules, record payments, track outstanding balances and due dates.	High
8	 Notifications & Alerts	The system shall send real-time notifications and reminders (email, SMS, in-app) for applications, approvals, disbursements and repayments.	Medium
9	 Reporting & Dashboards	The system shall provide dashboards and reports for loans, repayments, delinquencies, customers, and business performance.	Medium
10	 Business Rules & Workflows	The system shall enforce business rules for eligibility, loan limits, interest rates, approval workflows, and exceptions.	High
11	 External Interfaces & Integrations	The system shall integrate with third-party APIs and services (Credit Bureau, KYC, Payment Gateway, Bank, SMS/Email).	High
12	 Compliance Management	The system shall maintain audit trails and ensure compliance with applicable laws, regulations, and data privacy policies.	High
13	 Data Management	The system shall capture, validate, store, update, and archive data accurately across all modules.	Medium
14	 Other Functional Requirements	Any additional functional requirements specific to Smart Lender will be added here.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	 Performance & Speed	The system shall respond quickly to user actions and process transactions efficiently.	e.g., Page loads in < 2 seconds, API response time < 500 ms
2	 Scalability	The system shall handle growth in users, data, and transactions without degradation.	e.g., Supports up to 10,000 concurrent users
3	 Security & Data Privacy	The system shall protect data from unauthorized access, breaches, and threats.	e.g., End-to-end AES-256 encryption, Secure authentication, OWASP Top 10 compliance
4	 Reliability & Availability	The system shall be reliable and available for users as per business needs.	e.g., 99.9% uptime per month
5	 Usability & Accessibility	The system shall be easy to use and accessible to all intended users.	e.g., Complies with WCAG 2.1 AA standards
6	 Data Integrity	The system shall ensure accuracy, consistency, and completeness of data.	e.g., Zero data loss, ACID compliance for transactions
7	 Maintainability	The system shall be easy to maintain, update, and support.	e.g., Modular code, documented APIs, < 2 hours to deploy minor changes
8	 Portability	The system shall run on supported platforms, browsers, and devices without major changes.	e.g., Supports latest 2 versions of major browsers and mobile devices
9	 Compliance	The system shall comply with legal, industry, and organizational standards.	e.g., GDPR, HIPAA (where applicable), RBI guidelines, KYC/AML regulations
10	 Backup & Recovery	The system shall ensure data backup and recovery in case of failures or disasters.	e.g., Daily backups, RPO < 15 min, RTO < 1 hour
11	 Monitoring & Logging	The system shall log important events and monitor performance and errors.	e.g., Real-time monitoring, alerts for critical issues, logs retained for 90 days
12	 Other Non-Functional Requirements	Any additional non-functional requirements specific to Smart Lender will be added here.	e.g., Sustainability, Localization, Third-party SLAs, etc.